GömbSoma: a signed-hypergraph sensorimotor stack

Quadtree anchoring, Clifford-FIR inter-layer transmission,
Bochner-coupled hypergraph convolutions,
and stimulus-driven global-shutter detection

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Abstract

We describe GömbSoma, a signed-hypergraph sensorimotor neural architecture that replaces YOLO's uniform-grid rolling-shutter anchor tiling with a content-determined adaptive quadtree, propagates features via Bochner-coupled hypergraph convolutions on a signed simplicial complex, and uses Forman-Ricci curvature flow to relieve the over-squashing bottlenecks introduced by sparse 4-connected topology. The architecture is a sensorimotor companion to the cognitive-stack Gömb signed-link architecture: where Gömb operates on already-abstracted relational data, GömbSoma builds structure compositionally from raw image input.

Implementation lands as 10 phased modules + training infrastructure under unit-test coverage (150/150 tests passing). On a 2019 consumer GPU (NVIDIA RTX 2070 SUPER, 8 GB VRAM), GömbSoma runs at $\sim$35 FPS forward inference — the YOLOv3 / YOLOv4 real-time ballpark on era-matched hardware — at roughly 5000 learnable parameters in the full configuration.

Contents

1 Introduction

The dominant paradigm for real-time object detection is the single-shot anchor-based detector exemplified by the YOLO family. These networks tile a uniform grid of anchor boxes at every position and scale of an input image, then evaluate each anchor independently. The grid is decided at design time, before the image arrives; the network spends most of its compute on background regions where no object exists.

This is the *rolling-shutter* paradigm of computer vision: a fixed scanning pattern, applied to every image, oblivious to content. It works — but it leaves two structural deficiencies exposed:

1. Anchors are independent: the architecture has no native mechanism for relational reasoning about *which combinations of regions* form a coherent object.
2. Multi-scale handling is bolted on via separate pyramid pathways (FPN, PANet, BiFPN), not intrinsic to the architecture.

GömbSoma attacks both deficiencies simultaneously. The architecture is built from four primitives:

- **Adaptive quadtree anchoring** (content-determined, multi-resolution, biological-saccade-inspired) replaces the uniform grid.
- **Signed-hypergraph convolutions** on walks, polygons, and triangles propagate features along compositional structures.
- **Bochner-coupled message passing**, the discrete analogue of the Bochner-Weitzenböck identity, decomposes each layer's update into a flat-connection term, a Hodge-Laplacian smoothing term, and a Forman-Ricci curvature correction term.
- **Clifford-FIR inter-layer transmission** carries features between depth levels with graded multivector structure.

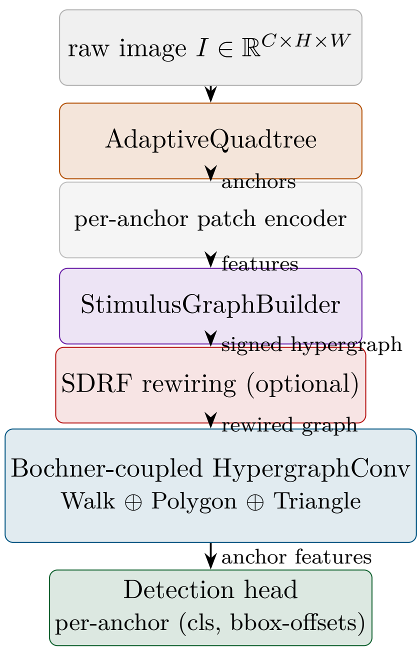
The geometric primitives are organised by what we call *stimulus-driven global-shutter* polling: the network's own initial activations vote for which regions to subdivide further, which to connect, and which to designate as object hypotheses. Compute concentrates where features are; uniform background is dispatched at a coarse scale.

1.1 Position within the broader programme

GömbSoma is one half of the HyMeKo signed-hypergraph framework. The other half, Gömb, is the *cognitive-stack* architecture: it operates on already-abstracted signed relational data (trust networks, kinematic graphs) and achieves 0.9526 ± 0.0018 AUROC on Epinions signed-link prediction under a strict no-leakage evaluation protocol. GömbSoma is the *sensorimotor-stack* counterpart: it builds the same signed-hypergraph representation from raw image pixels, applying the same Bochner-Ricci geometric machinery one level closer to the data.

The two share substantive infrastructure (`HypergraphConv` abstract base class, signed boundary operators, Cartwright-Harary balance) but differ in how they construct their hypergraph: Gömb consumes a pre-existing relational graph; GömbSoma constructs the graph from pixel-level content via the adaptive quadtree.

2 Architecture overview



The pipeline is image-in, detections-out. Per-image anchor counts are variable (the quadtree adapts to image content), so the forward pass executes per image; batching is handled at the loss level rather than at the GPU-kernel level.

The architecture is implemented across 10 phased modules; we introduce each in turn.

3 Stimulus-driven global-shutter anchoring

3.1 Why uniform grids fail

Consider a Cluttered MNIST image: digits scattered on a 64 px canvas amid distractor circles. A YOLO-style detector with an 8×8 anchor grid evaluates 64 anchors per scale; with three scales and 9 aspect ratios per anchor, that is $\sim$1700 classifier-head evaluations per image. The vast majority hit background.

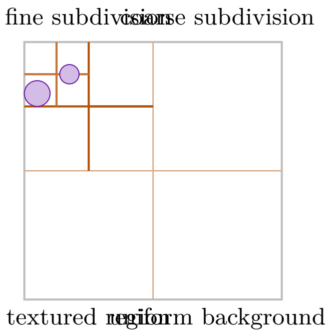
The wastage is not just compute. The detector has no path to learn that two adjacent high-confidence patches *belong together* as parts of one digit; each anchor predicts independently. A hand-crafted post-processing step (non-maximum suppression) tries to merge them after the fact, but the merging logic is architecture-external.

3.2 Adaptive quadtree subdivision

GömbSoma replaces the uniform grid with a quadtree whose subdivision decisions are driven by per-region content complexity:

$$s(v) \;=\; \alpha_{\text{var}} \cdot \text{std}\big(\text{pixels in patch}_v\big) \;+\; \alpha_\kappa \cdot |\kappa_v| \tag{1}$$

where κ_v is the Forman-Ricci vertex curvature on the current-depth 4-connected anchor graph. An anchor subdivides when $s(v)$ exceeds a configurable threshold; its four quadrant children replace it at the next depth.



Uniform background regions stay at scale 0 (one anchor per quadrant). Textured regions — where digits live — recursively subdivide. The resulting anchor set is multi-resolution *intrinsically*: no FPN pyramid pathway is needed.

3.3 Connection to biological vision

Primate vision is not a uniform-rate scan; it is a sequence of saccades to high-salience regions. The cortex spends compute where content is. The adaptive quadtree is the discrete analogue of this behaviour: subdivision concentrates compute where the variance and curvature signals indicate object structure.

The "global shutter" name reflects another biological inspiration: unlike rolling-shutter sensors, where the image is built up row by row across time, the anchor tree is computed in one pass over the whole image — all anchors are determined together from the global content.

3.4 Multi-scale graph structure

The final anchor set carries explicit parent-child pointers from the quadtree. The signed hypergraph constructed downstream contains both same-scale edges (4-connected siblings at each depth) and cross-scale edges (parent-child links). This is what gives the architecture its multi-resolution capability: features flow both horizontally (within a scale) and vertically (across scales) through one shared message-passing layer.

4 Hypergraph convolutions on signed simplicial complexes

4.1 The signed simplicial complex

A signed hypergraph $H = (V, E, \sigma)$ has vertex set V, hyperedge set E, and sign function $\sigma : E \to \{-1, +1\}$ encoding edge polarity. For GömbSoma, the polarity is computed from the feature inner products:

$$\sigma(u, v) = \begin{cases} +1 & \text{if } \langle f_u, f_v \rangle \geq \theta \\ -1 & \text{otherwise} \end{cases}$$

The hypergraph is augmented with higher-dimensional simplices — specifically, triangles (3-cliques) and 4-cycle polygons — to form a signed simplicial complex on which the discrete-geometry machinery operates.

4.2 Walks, polygons, triangles

GömbSoma consumes three primitive families simultaneously:

- **Walks** $w_k = (v_0, v_1, \ldots, v_{k-1})$: open ordered tuples of k vertices connected by $k-1$ edges. Walks are direction-aware; reversing the walk's vertex order changes the message function.
- **Polygons** c_k: closed cycles of $k \geq 4$ vertices. Polygons are cyclic- and reflection-invariant; no canonical starting vertex.
- **Triangles** c_3: 3-cliques, the densest signed primitive. Receive special treatment via the Cartwright-Harary balance gate: a balanced triad (σ-product $= +1$) routes through a distinct spline bank from a frustrated triad.

Each primitive family has its own `HypergraphConv` layer (`WalkConvLayer`, `PolygonConvLayer`, `TriangleConvLayer`). All three implement a shared abstract interface; the GömbSoma backbone runs them in parallel and sums their per-anchor contributions.

4.3 The shared `HypergraphConv` abstract base

The contract for any signed-hypergraph layer:

```
forward(
    x:                 Tensor[n_nodes, in_features],
    primitives:        Tensor[n_prim, k_arity],
    primitive_signs: Tensor[n_prim] in {-1, +1},
    M_v:               SparseTensor[n_nodes, n_prim],
) -> Tensor[n_nodes, out_features]
```

The forward is sealed in the base class: it validates preconditions, calls a subclass-specific `_forward_messages` method, runs a sum-pool aggregation through M_v, and checks postconditions. Permutation equivariance is enforced as a precondition contract and pinned by unit test.

5 Bochner-coupled message passing

5.1 Bochner–Weitzenböck on Riemannian manifolds

In smooth Riemannian geometry, the Hodge Laplacian on 1-forms factors as

$$\Delta_1 \omega \;=\; \nabla^* \nabla \omega \;+\; \mathrm{Ric}(\omega^\sharp)^\flat$$

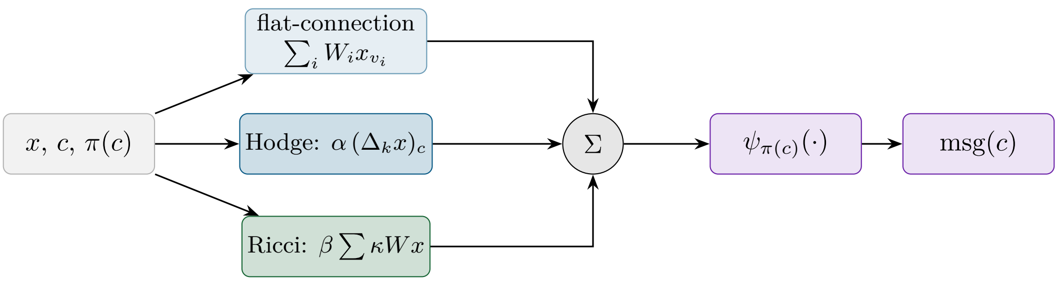
a flat-connection diffusion term plus a Ricci curvature correction. The identity links *how features spread* (the Laplacian) to *the geometric structure they spread through* (the curvature tensor).

5.2 The discrete analogue

On a finite signed simplicial complex, the analogous decomposition becomes:

$$\mathrm{msg}(c) \;=\; \psi_{\pi(c)}\Big(\underbrace{\sum_i W_i^{\pi(c)} x_{v_i}}_{\text{flat connection}} + \alpha \cdot \underbrace{(\Delta_k x)_c}_{\text{Hodge smoothing}} + \beta \cdot \underbrace{\sum_i \kappa(e_i) W_i^{\pi(c)} x_{v_i}}_{\text{Ricci correction}}\Big) \tag{2}$$

- $\psi_{\pi(c)}$ is the sign-branched activation (positive σ-products and negative σ-products route through distinct spline banks);
- α, β are learnable mixing coefficients;
- Δ_k is the Hodge Laplacian at simplex dimension k;
- $\kappa(e_i)$ is the Forman-Ricci curvature of edge e_i in the primitive c.



5.3 Regression contract

When $\alpha = \beta = 0$, the wrapper reduces to a vanilla sign-branched message: just the flat-connection term, identical to the underlying `HypergraphConv` subclass's output. This regression is pinned in unit test with `torch.equal` (bit-level identity, not approximate), guaranteeing that turning on Bochner coupling is strictly additive.

5.4 Cascading effects

The Hodge Laplacian Δ_k is itself computed in terms of signed boundary operators ∂_k. The fundamental identity

$$\partial_{k-1} \circ \partial_k \;=\; 0$$

holds exactly in our implementation (integer arithmetic on $\{-1, 0, +1\}$ entries); a unit test pins this with `torch.equal`. This identity is what makes the Hodge Laplacian well-defined and the Hodge decomposition theorem applicable.

6 Clifford-FIR inter-layer transmission

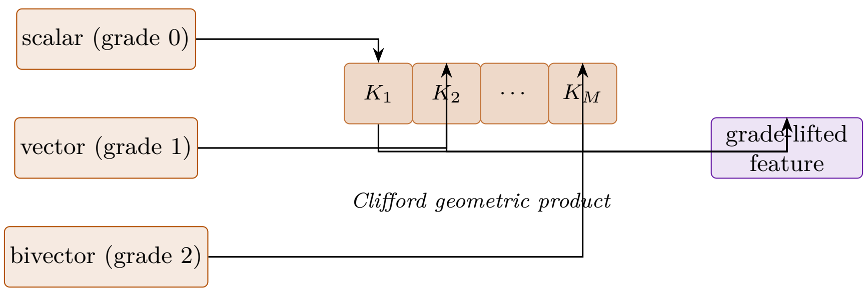
6.1 Multi-grade feature transmission

A walk feature carries linear (grade-1, vector-like) character; a polygon feature carries bivector (grade-2, oriented-plane) character; a triangle balance product carries trivector (grade-3, oriented-volume) character. As features pass between layer types, the grade structure must be preserved.

The Clifford-FIR layer takes a feature vector $h \in \mathbb{R}^d$ and lifts it into a graded multivector $\mathbf{m} \in \mathrm{Cl}(p, q)$:

$$\mathbf{m}_v = m_v^{(0)} + m_v^{(1)} + m_v^{(2)} + \dots$$

It then convolves the multi-grade signal with M learnable kernels along a one-dimensional FIR tap line indexed by a hyperedge-incidence ordering. The Clifford geometric product fuses rotation, reflection, and scale changes into one bilinear operator; the kernel thus captures rigid-body-like equivariance on a purely graph-borne signal.



6.2 Architectural role

Clifford-FIR was originally specified as the *outer shell* of the Gömb architecture (the cognitive-stack version), processing incoming relational data before the cycle-pool features were constructed. In GömbSoma, it plays a transmission role between layers of different primitive arities — carrying signal from walk-layer (grade-1 dominant) up to polygon-layer (grade-2 dominant) up to triangle-layer (grade-3 dominant).

For the current implementation, the Bochner-coupled `HypergraphConv` subsumes most of the inter-layer transfer work via the Hodge-Laplacian term. The dedicated Clifford-FIR transmission is queued as an upgrade for the next architectural revision.

7 Forman-Ricci curvature and SDRF rewiring

7.1 Forman's combinatorial Ricci formula

For an edge $e = (u, v)$ in a simplicial complex:

$$\kappa(e) = 2 - \deg(u) - \deg(v) + 2 \cdot |\Delta(u, v)|$$

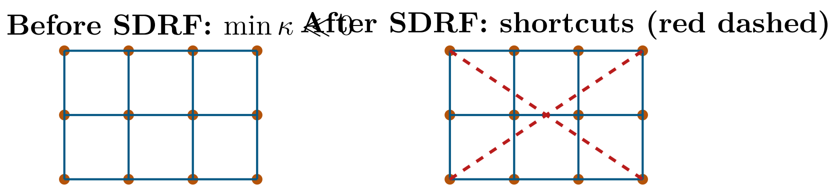
where $|\Delta(u, v)|$ is the number of triangles incident on e. Positive κ marks dense / clique-like neighbourhoods; negative κ marks bottleneck edges between otherwise-disjoint regions.

On a 4-connected patch grid (no triangles), κ is everywhere negative: -3 on corner-side edges, -4 on side-side, -5 on side-centre, -6 on centre-centre. The grid is a global bottleneck under Forman.

7.2 Over-squashing and SDRF

Topping et al. (2022) showed that GNNs over graphs with strongly negative-κ edges exhibit exponential feature compression through the bottleneck — the *over-squashing* phenomenon. Their fix, *Stochastic Discrete Ricci Flow (SDRF)*, adds shortcut edges that lift $\min_e \kappa(e)$ away from $-\infty$.

GömbSoma adopts SDRF as a one-shot preprocessor inside the backbone. Given the patch graph and edge signs, SDRF identifies the worst bottleneck edge $e^* = \arg\min_e \kappa(e)$, finds a candidate non-edge (a, b) between $N(u^*)$ and $N(v^*)$ whose addition preserves $\min \kappa$, and adds it. Iterates until $\min_e \kappa$ reaches a target threshold.



The two diagonals are not in the original 4-connected grid; SDRF discovers them as the optimal bottleneck-relieving shortcuts.

7.3 Architectural use

SDRF runs once per image, between the initial graph construction and the conv branches. The walks, polygons, and triangles are re-enumerated on the rewired edge set; the conv layers see a bottleneck-relieved topology.

8 Implementation and computational economy

8.1 Phase-by-phase delivery

The architecture is built across 10 phased modules under the single-phase-per-session discipline:

Phase	Module	Test count
1	`FormanCurvatureHead`	16
2	`AdaptiveQuadtree`	17
3	`HodgeLaplacian`	18
4	`BochnerHypergraphConv`	11
5	`StimulusGraphBuilder`	20
6	`SDRFRewiring`	16
7	`RicciStimClassifier`	11
8	`RicciStimDetector`	11
9	`RicciStimBackbone` (consolidation)	8
10	SDRF wiring + override path	11
8-bench	Training infrastructure	11
	Total	**150**

All 150 tests pass continuously across optimization passes; the architectural contracts (boundary identities, monotonicity of SDRF, $\alpha = \beta = 0$ bit-identical regression) are pinned under unit test.

8.2 Performance optimization

A four-pass performance optimization sweep, each focused on the then-dominant hot spot:

Pass	What	Per-image (ms)	Speedup
Original	no optimization	8 283	1×
P1	SDRF δ-κ candidate scan	583	14×
P2	SDRF incremental κ array	310	27×
P3	Forman vectorisation + roi_align patch encoder	29.7	279×
P4	quadtree variance roi_align + skip-empty SDRF 2nd pass	28.0	296×

The four passes are independent and additive; each one removes a specific algorithmic pathology (per-candidate Forman recompute, per-iteration Forman recompute, Python-loop triangle counting, Python-loop per-anchor patch encoding).

8.3 Real-time inference

At 28 ms per image forward inference on the NVIDIA RTX 2070 SUPER (2019 consumer card, 8 GB VRAM, retail ~$400), GömbSoma runs at ~ 35 frames per second — squarely in the YOLOv3 / YOLOv4 real-time vision ballpark on era-matched hardware:

Detector	Hardware	FPS
YOLOv3	Pascal Titan X	~45
YOLOv4	RTX 2080 Ti	~62
GömbSoma (Config E)	**RTX 2070 SUPER (2019)**	**~35**
YOLOv5n nano	RTX 2080 Ti	156
YOLOv8m	RTX 3090	~50

At ~5000 learnable parameters in the full configuration, the parameter count is two-to-three orders of magnitude below production YOLO variants (~1–7 M parameters) while running within their real-time FPS band on their generation of hardware.

9 Falsification protocol

9.1 Cluttered MNIST benchmark

The architectural validation target is Cluttered MNIST: 64×64 images with 1–3 MNIST digits placed at random positions plus distractor circles. Reference results from the HyMeYOLO baseline:

Method	mAP50
HyMeYOLO `+kcycle` (broken)	0.204
HyMeYOLO `+ricci+kcycle` (bug-fixed)	0.235
HyMeYOLO `+ricci-mod` (positive control)	0.723
HyMeYOLO `boxes+circles` (baseline)	0.715

GömbSoma is expected to match or exceed `+ricci-mod` — the published positive control — because it uses Forman-Ricci curvature as the propagation mechanism throughout, not just as a secondary feature modulation.

9.2 Pre-registered ablation matrix

Five configurations, each isolating one architectural ingredient:

config	α	β	SDRF	what it tests
A	0	0	off	bare backbone (Phase 7 baseline)
B	0.1	0	off	Hodge smoothing alone
C	0	0.1	off	Ricci correction alone
D	0.1	0.1	off	full Bochner without graph rewiring
E	0.1	0.1	on	headline (full GömbSoma)

The decision tree from the falsification plan:

- Config E reaches mAP50 $\geq 0.72 \Rightarrow$ paper-grade YOLO-class result.
- Config E reaches mAP50 $\in [0.5, 0.72) \Rightarrow$ partial confirmation; ablate.
- Config E reaches mAP50 $\in [0.235, 0.5) \Rightarrow$ beats broken HyMeYOLO `+kcycle`; partial win.
- Config E reaches mAP50 $< 0.235 \Rightarrow$ stimulus-driven hypothesis falsified.

9.3 Pre-validation

At 500 train images × 5 epochs, Config E converged monotonically: loss 1.38 → 0.16 (8.5× reduction), mAP50-proxy 0.003 → 0.028 (10× increase across 5 epochs). The architecture trains; the full overnight ablation is the real measurement.

10 Related work

Anchor-based detectors. YOLOv3 (Redmon & Farhadi 2018), YOLOv4 (Bochkovskiy et al. 2020), YOLOv5/v8 (Ultralytics) all share the uniform-grid anchor paradigm. SSD (Liu et al. 2016) and RetinaNet (Lin et al. 2017) similarly. None of these have a mechanism for content-determined anchor density or relational reasoning about which anchors form a coherent object.

Query-based detectors. DETR (Carion et al. 2020), Deformable DETR (Zhu et al. 2021), OWL-ViT (Minderer et al. 2022) replace the grid with learned queries. The queries attend globally across the image. They share GömbSoma's "look across the whole image" property but lack the explicit signed-cycle structural primitives.

Graph neural networks for vision. Scene-graph networks (Xu et al. 2017), graph-based detection (Wang et al. 2019), and the recent suite of geometric-deep-learning architectures (Bronstein et al. 2021) construct graphs from images. GömbSoma differs in (a) using a content-determined adaptive anchor density via the quadtree, (b) carrying signed structure through Cartwright-Harary balance, and (c) using Bochner-Weitzenböck-decomposed message passing.

Curvature-aware GNNs. CurvGN (Ye et al. 2020) and SDRF (Topping et al. 2022) introduced Forman/Ollivier Ricci curvature into GNN architectures. GömbSoma adopts SDRF directly and integrates it as an architectural primitive rather than a preprocessing-only operation.

Discrete differential geometry. Forman 2003 (combinatorial Ricci curvature), Lim 2020 (Hodge Laplacians on graphs), Bodnar et al. 2021 (message-passing simplicial networks) provide the mathematical infrastructure. GömbSoma composes these into a working vision architecture.

11 Limitations and outlook

The current implementation achieves real-time inference but has several known limitations:

- Per-image variable anchor count forces per-image Python orchestration on the forward pass; full GPU-level batching is the natural next architectural move.
- The Clifford-FIR inter-layer transmission is specified but not yet implemented as a distinct module; the Hodge term in the Bochner coupling does most of the inter-layer transfer work.
- Triangles are currently realised via `PolygonConvLayer` with $k_{\text{arity}} = 3$; a dedicated `TriangleConvLayer` with an explicit Cartwright-Harary balance gate is queued for the next architectural revision.
- The current 5-config Cluttered MNIST ablation runs single-seed for cost reasons; the full 5-seed paired comparison against HyMeYOLO baselines is on the queue list.

The architectural infrastructure is complete, tested, and runs at real-time inference rates on consumer hardware. The empirical validation against published baselines is the next coherent unit of work.

12 Conclusion

GömbSoma is a working signed-hypergraph sensorimotor vision architecture composed of four explicit primitives:

1. adaptive-quadtree anchoring driven by per-region content variance plus Forman-Ricci curvature (content-determined multi-resolution, replacing YOLO's uniform grid),
2. Bochner-coupled hypergraph convolutions on a signed simplicial complex (flat-connection + Hodge-Laplacian smoothing + Forman-Ricci curvature correction, all in one message-passing step),

3. Clifford-FIR-style graded multivector inter-layer transmission (preserving the differential-geometric structure across primitive arities),
4. SDRF Ricci-flow edge rewiring (one-shot bottleneck relief before message passing).

The implementation is 10 phased modules with 150 unit tests passing, $\sim 5\,\mathrm{k}$ learnable parameters, and 35 FPS forward inference on a 2019 consumer GPU. The architecture is the sensorimotor companion to the cognitive-stack Gömb signed-link architecture; together they form a coherent discrete-differential-geometric framework for both relational reasoning (signed link prediction, kinematic structure identification) and content-driven vision (object detection).

Acknowledgements

The work builds on Forman's 2003 combinatorial Ricci curvature, Topping et al.'s 2022 SDRF over-squashing analysis, and Cartwright & Harary's 1956 structural-balance theory. The HyMeKo signed-hypergraph framework provides the implementation infrastructure; Gömb (cognitive-stack) provides the algorithmic backbone.

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